

# ADP AI - Prediction Report

Generated by ADP AI Random Forest Prediction Pipeline

Report ID: f39a73c9-7fc5-474c-a972-3a5f7676159f

Algorithm: Random Forest Regressor

Model Type: regression

Accuracy Score: 99.42%

## Dataset Details

Target Column: Profit

Total Rows: 500

Total Columns: 5

## Model Selection

AI Detected Intent: auto

Final Task Type: regression

Decision Reason: The target column is numeric and continuous, so ADP AI selected Random Forest Regressor.

## Metrics

R2 Score: 0.9942

Mean Absolute Error: 9659.8164

Root Mean Squared Error: 11943.9113

Training Rows: 400

Testing Rows: 100

## Top Feature Importance

Revenue: 0.593549

Expenses: 0.401798

Assets: 0.00239

Liabilities: 0.002263

## Prediction Samples

Actual: 87685.0 | Predicted: 79332.1367

Actual: 411820.0 | Predicted: 415916.7

Actual: 2905.0 | Predicted: -1128.5333

Actual: 172144.0 | Predicted: 183117.77

Actual: 339992.0 | Predicted: 317709.47

Actual: -72470.0 | Predicted: -65651.3167

Actual: 119648.0 | Predicted: 121108.6567

Actual: 292589.0 | Predicted: 278040.45

Actual: 197080.0 | Predicted: 190994.54

Actual: 165635.0 | Predicted: 160430.7

## Smart Recommendations

1. Data quality: No major missing values found in the uploaded dataset.
2. No duplicate rows found, so the dataset is suitable for prediction analysis.
3. Target analysis: Average Profit is 52748.32, median is 56517.0, minimum is -338246.0, and maximum is 451677.0.
4. High priority: Improve 'Revenue' because it has a positive relationship with Profit.

Random Forest importance is 59.35%.

5. Control 'Expenses' because higher values may reduce Profit. Correlation detected is -0.615.

6. Control 'Assets' because higher values may reduce Profit. Correlation detected is -0.005.

7. High priority: Improve 'Liabilities' because it has a positive relationship with Profit. Random Forest importance is 0.23%.

8. Improvement plan: Focus on the top important features that influence 'Profit' because they have the strongest impact on prediction.

## Prediction Flow

1. User uploaded dataset
2. ADP AI read rows and columns
3. ADP AI checked target column
4. ADP AI cleaned missing values
5. ADP AI prepared numeric and categorical features
6. ADP AI selected suitable ML model
7. Random Forest model trained using train-test split
8. Model predictions generated
9. Evaluation metrics calculated
10. Feature importance generated
11. Recommendation intelligence generated
12. Prediction report prepared